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Price and subsidy under uncertainty: Real-option approach to optimal investment decisions on energy storage with solar PV

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Abstract

Renewable generation sources still have not achieved economic validity in many countries including Korea, and require subsidies to support the transition to a low-carbon economy. An initial Feed-In Tariff (FIT) was adopted to support the deployment of renewable energy in Korea until 2011 and then was switched to the Renewable Portfolio Standard (RPS) to implement more market-oriented mechanisms. However, high volatilities in electricity prices and subsidies under the RPS scheme have weakened investment incentives. In this study we estimate how the multiple price volatilities under the RPS scheme affect the optimal investment decisions of energy storage projects, whose importance is increasing rapidly because they can mitigate the variability and uncertainty of solar and wind generation in the power system. We applied mathematical analysis based on real-option methods to estimate the optimal trigger price for investment in energy-storage projects with and without multiple price volatilities. We found that the optimal trigger price of subsidy called the Renewable Energy Certificate (REC) under multiple price volatilities is 10.5% higher than that under no price volatilities. If the volatility of the REC price gets doubled, the project requires a 26.6% higher optimal investment price to justify the investment against the increased risk. In the end, we propose an auction scheme that has the advantage of both RPS and FIT in order to minimize the financial burden of the subsidy program by eliminating subsidy volatility and find the minimum willingness-to-accept price for investors.

Keywords

Renewable energy, energy storage system, renewable portfolio standard, feed-in-tariff, price uncertainty, subsidy uncertainty, real option analysis

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Introduction

With increasing pressure to reduce carbon emissions and promote sustainable energy sources, the transition to renewable energy has become a top agenda in many countries. As one of those countries, Korea also announced the 'Energy Transition Policy' in 2017, the main goal of which is to convert an energy system oriented toward nuclear and coal power to one oriented toward renewable sources and to set up the roadmap to deploy renewable generation up to 20% of total energy consumption by 2030. Among the 20% renewable generation, solar photovoltaic (PV) and wind power are the main sources. They are scheduled to be deployed up to 33.4 GW and 17.7 GW, respectively, by 2030, compared to the current status of approximately 13 GW and 1.5 GW in 2020.[1] In addition, the government has announced a more ambitious roadmap to deploy renewable sources up to 30~35% of total power consumption by 2040.[2]

However, as the share of variable renewable sources increases, the reliability and stability of the power supply get undermined because of the high variability and uncertainty of those renewable sources. In order to mitigate the variability and uncertainty caused by renewable sources, various backup resources need to be considered to support variable renewable sources and energy storage is one of the most likely resources. Energy storage can reduce the variability of renewable sources by moving peak demand to off-peak hours, especially filling the valley during the daytime, so called 'duck curve', mainly caused by high solar generation. Also, energy storage can mitigate problems caused from uncertainty by providing operating reserves which are needed to securely balance the power system in real time. Recently, these costs that variable renewable sources cause in the power system are called the 'integration cost', that is, the cost incurred when integrating renewable sources in the power system; it consists of profile cost, balancing cost, and grid cost. In this perspective of the integration cost, energy storage is an effective resource to reduce profile cost and balancing cost mainly caused by variability and uncertainty, respectively.

In order to promote the deployment of conventional renewable sources, such as solar PV and wind, and non-conventional sources, such as energy storage, many countries exercise subsidy programs. Korea also has approximately 20 years of experience with subsidy programs for renewable sources. The Feed-In Tariff (FIT), introduced in 2002, guarantees a fixed profit securely for a long time in order to foster renewable sources in their early stage, when they are far from achieving economic feasibility. In 2012, FIT was replaced by the Renewable Portfolio Standard (RPS), which is a more market-based program that provide a variable subsidy called the Renewable Energy Certificate (REC), whose value is determined by the law of supply and demand in the real-time competitive market like a stock market.

The characteristics of FIT and RPS can be highlighted as follows. The advantage of FIT is that it can help promote the renewable source in the early stage, because it provides a secure subsidy. Its disadvantage is that searching for a reasonable subsidy level is challenging, and when it is overestimated, the government's financial burden increases and the incentive for investors to reduce the cost of renewable-energy projects weakens. The advantage of the RPS is that it creates a strong incentive to reduce the cost of a renewable-energy project because of the competitive market scheme, but its disadvantages are that the projects are exposed to uncertainties of both electricity price and REC price, which make the financial cost of the project increase, which increases renewable-energy costs eventually.

In the current RPS framework in Korea, energy storage acquired 5 REC in 2019 compared to 1~1.5 REC for solar PV and wind-generated power, per MWh of renewable energy produced. Hence energy storage receives a subsidy approximately 5 times that of other conventional renewable sources, which shows that the importance of energy storage is well acknowledged and that the government strongly promotes energy storage to support various renewable sources in the power system. However, dual uncertainties from electricity prices and the REC price significantly discourage investors to initiate projects for energy storage. Hence it is crucial to understand how the current high price uncertainties will affect investment decisions and timing under the RPS program in order to evaluate whether the current subsidy program will be able to achieve its ambitious target for renewable sources.

In this study, we analyzed the optimum trigger price of investment for energy storage projects with dual uncertainties under the RPS program and also estimated the optimum investment price without REC price uncertainty, which can be considered to be an FIT program. Comparisons of the two results can show how the two most widely adopted policies for renewable energy promotion have different effects on the optimal investment price and how the price uncertainty plays a key role. For a methodology, we applied theoretical analysis based on a real-option method with a value function reflecting the characteristics of the energy storage profit structure. We assumed two independent processes of Geometric Brownian Motion (GBM) for electricity price and REC price in order to reflect the dual price uncertainties realistically. We derived an analytic solution to find the optimal investment price, and applied Korean market data to compute the optimal investment price level under RPS and FIT.

The structure of this paper is as follows. The next section discusses relevant literature and Then, the theoretical model used in this study are proposed and developed. 'Numerical example' section estimates the optimal threshold prices for investment in each scenario based on Korean data, and in the penultimate section conducts sensitivity tests to see the effect of the main variables on optimal investment prices. In the final section, policy implications for subsidy programs based on the results derived are discussed.

Literature review

Studies such as Pindyck,[3] Heydari[4] and Walsh[5] applied real option methodology to analyze the environmental policy or technologies under various market and policy uncertainties. Pindyck[3] analyzed the optimal timing problems for environmental policies by considering two uncertainty factors which are uncertainty over environmental change and uncertainty over the social cost of environmental damage. This study used mathematical approach by applying two types of real option methods, two-period model and continuous-time model.

Heydari[4] estimated the option value of adding co2 capture and sequestration(CCS) to existing coal-fired power plant under the uncertainty of electricity price, CO2 price and coal price. It analyzed two technologies, full CCS(FCCS) and partial CCS(PCCS) and showed when CO2 price is high, FCCS is more economically valid and the optimal boundary condition is largely dependent on CO2 price volatility. Walsh et al.[5] proposed two mathematical models for the optimal investment decision in CCS technology, one with deterministic carbon price and one with stochastic price. This study showed that investment cost of CCS is time dependent, and reduction of carbon price volatility by adopting carbon tax scheme or carbon floor policy in U.K will make CCS technology more viable option in near future.

Several literatures such as Lin and Wesseh,[6] Lee et al.[7] and Bockman et al.[8] analyzed the investment value of renewable generation sources under uncertainties of electricity price or subsidy program. Lin and Wesseh[6] used a real option pricing approach to evaluate the investment value of solar power technology under uncertainty of fossil fuel prices when feed-in tariff (FIT) policy is imposed in China. It used simulation method and showed the FIT level should be increased to achieve the optimal point where more investment incentive is achieved with minimal government financial burden.

Lee et al.[7] evaluated the investment value of R&D for wind power technology and the optimal investment timing using a binomial tree approach. This paper assumed that a decision maker can make a choice to either abandon, deployment or continue the R&D, and the empirical result showed that considerable investment value exists in continuing wind power R&D in Korea. Bockman et al.[8] applied real option method with continuous scaling to evaluate the investment value of small hydropower projects under the uncertain electricity prices. The analysis is conducted based on the simulation approach based on a spreadsheet model.

A literature like Henao et al.[9] estimated the economic value of providing flexibility in transmission expansion planning in the power system based on the real option approach with binomial trees. The paper argued that traditional DCF method disregard the responses from the planner when more new information is available and uncertainty is reduced, and the planner may want to expand or defer based on those information. The study showed that incorporating flexibility in transmission expansion planning is beneficial and increase social welfare.

There are studies analyzing empirical linkage among renewable generation deployment, environmental sustainability and economic development. Quantifying these linkages is important when developing the subsidy policies for renewable generation as this provides evidence to justify such support. Ahmad et al.,[10] Alvarado et al.,[11] Ahmad et al.[12] and Fatima et al.[13] analyze the relationship between deployment of renewable generation, CO2 emission and economic development and explore the possibility of sustainable growth based on green energy. Ali et al.[14] and Jabeen et al.[15] estimated factors affecting economic validity of renewable sources such as OFF-Grid solar PV and biogas widely deployed in the developing countries. Ahmad et al.[16] investigated how Haven/Halo hypothesis and EKC hypothesis have heterogeneous impact on environmental sustainability depending on the development level of provinces in China .

Many literatures discussed in this section focus to analyze the investment feasibility of carbon reduction technologies such as CCS or retrofit of existing coal-powered plant under carbon price volatility.[17] There exists some literatures analyzing investment feasibility of solar PV or wind generation under single SMP volatility, but not many studies analyzed investment feasibility of energy storage with thorough consideration of multiple volatility of SMP and REC price.[18]

This study focus on energy storage technology collocated with solar power farms, which can contribute to effectively mitigate the reliability problem caused by variability and intermittency of renewable sources in the power system.[19] Two uncertainties in electricity price and REC price are considered, and estimated optimal investment price for varying volatility levels for REC price. Finally, this study compares the optimal investment price of energy storage between RPS and FIT, and discusses the additional financial burden that government has to take under RPS scheme.[20,21]

The contribution that this paper provides from previous literatures can be summarized in three perspectives. Firstly, this paper applies a mathematical approach to find optimal trigger prices under multiple price volatilities based on real option approach and derives a closed-form solution under the RPS scheme in Korea. Secondly, this paper compares optimal trigger prices under RPS and FIT and provides implication to adopt more effective policy for the deployment of renewable sources. Thirdly, this paper offers sensitivity tests that affects optimal trigger prices such as drift rate or volatility of underlying price, REC multiplier and storage capital cost. This analysis provide important implication regarding how the policy design must be done to promote energy storage which is an essential resource to mitigate variability of renewable sources.

Model

In this study, we assume that investment has not been made at $t=0$. We let [I] be an indicator function for investment. when $[I]=0$, no investment has been made. Once the investment for energy-storage project is made as $[I]=1$, the project value can be illustrated as in equation (1).

$$\begin{aligned} V(P_{REC},\ P_{Gap}) &= E_0 \int_0^T \left(Q_{REC} \cdot P_{REC}(t) - Q_{Gap} \cdot P_{Gap}(t) - C_{o\&m}\right) e^{-rt} dt \\ &= E_0 \int_0^T \left(n \cdot E \cdot Q \cdot P_{REC}(t) - Q \cdot P_{Gap}(t) - C_{o\&m}\right) e^{-rt} dt \end{aligned} \tag{1}$$

The value function is composed of three components, profit from REC, cost from Gap price induced by charging at high prices and discharging at low prices, and cost from operations and management. $P_{REC}(t)$ is a variable market price of REC at time t. Q_{REC} is the number of RECs offered and can be computed by equation (2), where n is a REC multiplier, E is the energy-storage efficiency, and Q is energy initially purchased by storage.

$$Q_{REC} = nEQ \tag{2}$$

$P_{GAP}(t)$ is a variable price that is determined by the difference between charging and discharging prices at time t. The electricity price applied here is based on system marginal price (SMP), which is determined by a marginal generating unit that meets the demand in each hour. Under the RPS scheme in Korea, REC is offered based on the amount of energy shifted from nighttime to daytime in order to fill the valley caused by intense solar generation. Hence, P_{buy} is defined as the average price from 10 am to 4 pm, when solar generation is abundant, and P_{sell} is defined as the average price from 5 pm to midnight, when stored solar energy is mostly discharged back to the grid. When calculating P_{GAP}, one should consider the storage efficiency, E, because not all the energy purchased can be compensated for at P_{sell}, because there is energy loss during charging and discharging. Therefore, P_{GAP} can be understood as the gap price between the buy price and the sell price at which storage can make a profit, reflecting storage efficiency. Equations for $P_{GAP}(t)$ and P_{buy} are presented in equations (3) and (4).

$$P_{GAP} = P_{buy} - E \cdot P_{sell} \tag{3}$$

$$P_{buy} = \frac{\sum_{t=10}^{16} P_t}{7}; \quad P_{sell} = \frac{\sum_{t=17}^{24} P_t}{8} \tag{4}$$

$Q_{GAP} \cdot P_{GAP}(t)$ should be profit if buy price is lower than sell price, and this is true if the duck curve caused by solar generation is severe. However, in the current net load of Korea in 2019, daytime demand was still higher than nighttime demand so as SMPs. Therefore, $Q_{GAP} \cdot P_{GAP}(t)$ becomes a cost that storage needs to bear in order to receive RECs. Hence in equation (1), the value function is defined as the profit from REC and the cost from Gap price and O&M.

For simplicity, we will number the term related to REC as 1 and GAP as 2, as noted in equation (5)

$$P_1 = P_{REC}, \quad P_2 = P_{GAP} \tag{5}$$

Two prices in the value function follow a stochastic process, because they are volatile. We assume that the two price processes follow Geometric Brownian Motion (GBM) as shown in equations (6) and (7).

Energy price in long-run is known to follow mean-reverting process typically. However, according to Pindyck,[22] GBM can be a good approximation when the rate of mean reverting is low. Figure 1 shows the time-series graphs for SMP and REC for the study period. REC price clearly shows downward trend in recent years, which means that GBM is appropriate. The trend of SMP is not very clear but as discussed in Pindyck,[22] GBM can be a good approximation for energy price. In addition, SMP is expected to go down in the long run in Korea because when the share of renewable sources rapidly increase in the power system as planned, the marginal costs of renewable sources are zero, so they replace most expensive marginal generating units based on merit-order dispatch mechanism. This can lower overall SMP in the long-run, and support the application of GBM to SMP. The GAP price also shows a downward drift, because more solar PV capacity makes the Gap price continue to decrease by lowering electricity prices during peak hours. Hence, for these reasons, GBM is applied for processes of prices. In these equations, dw_t is a standard wiener process that

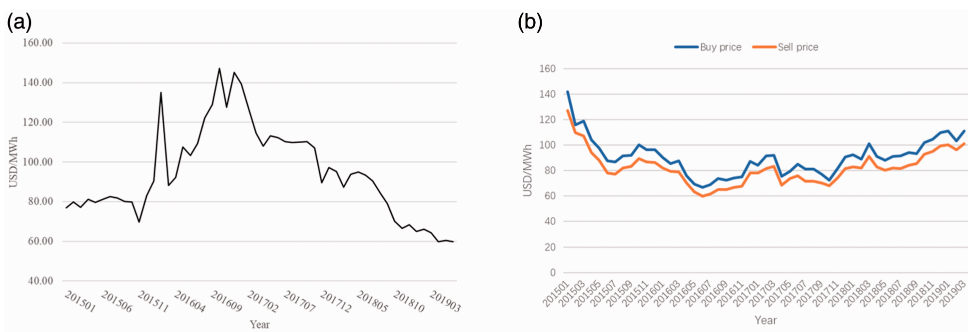

Figure 1. Time-series trend of REC price and SMP price (Buy price and Sell price). (a) REC price (b) Buy price and Sell price

satisfies $\mathrm{E}\left(dw_t^i\right) = 0, \mathrm{Var}(dw_t^i) = dt$. The correlation coefficient between the REC and GAP prices is ρ_{12}, which means $\mathrm{E}\left[\mathrm{dw}_t^1 dw_t^2\right] = \rho_{12}\mathrm{dt}$.

Here μ and σ show drift rate and volatility, respectively, and they are constant based on the market data.

$$\mathrm{dP_1} = \mu_1 \mathrm{P_1 dt} + \sigma_1 \mathrm{P_1 dw_t^1} \tag{6}$$

$$\mathrm{dP_2} = \mu_2 \mathrm{P_2 dt} + \sigma_2 \mathrm{P_2 dw_t^2} \tag{7}$$

where $\{\mu_\mathrm{i}; i = 1, 2\}$ and $\{\sigma_\mathrm{i}; i = 1, 2\}$ are drift and volatility parameter, respectively

Introducing two stochastic prices into the optimal investment problem makes the solution process very complicated, so we make two further assumptions that can simplify the mathematical derivation. First, we assume that the storage unit can be constructed and attached instantaneously to existing solar PV units. This assumption can allow the profit flows from both REC price and GAP price to be achievable at the moment of an investment decision. Also, we can see that the real-option approach to the investment decision on storage is appropriate, since the value of the option to defer the investment exists here, since once the investment decision is made, it is irreversible.

Second, we assume that the lifetime of storage is infinite. According to Few et al.,[23] Battery cycle life in 2020 is expected to be 6000 cycles to 13,500 cycles in the best scenario, which means that lifetime of LIB can be approximately 17 years to 37 years if one cycle is used per day as we assumed in this study. Few et al.[23] noted that LIB cost reduction will be limited as this side of technical improvement is now matured but life cycle improvement still has high potential and this will be the key to reduce per-cycle cost of LIB and validify the application of LIB to support variable renewable sources in the power system in the near future.

Infinite lifetime assumption is required to apply mathematical approach in simplifying the result from Bellman equation based on two GBM processes. It is much more challenging to solve a problem with finite horizon, and often closed-form solution cannot be derived and numerical approach becomes necessary. Similar previous studies such as Walsh et al.[5] and Heydari et al.[4] assumed infinite lifetime for coal-fired plants or carbon capture and storage (CCS) for this reason and stated that this is the limitation for the mathematical approach. However, we believe that this assumption has little impact on the solution because the discounting of future profit after 25 to 30 years will be greatly reduced and have very small contribution to the net present value of the project.

NPV of energy-storage project

Expected Net Present Value (NPV) of an energy-storage project can be derived as equation (8). E_0 is an expectation operator at $t = 0$, which is needed because the two prices have stochastic processes. The expected net present value with an infinite time horizon can be derived as in equation (8).

$$\begin{aligned} \mathrm{V}(\mathrm{P}_1, P_2) &= E_0\left[\int_0^\infty (\mathrm{nEQP_{1,t}} - \mathrm{QP_{2,t}} - \mathrm{C_{o\&m}}) e^{-rt} dt | P_{1,0} = P_1,\ \ P_{2,0} = P_2\right] \\ &= \frac{\mathrm{nEQP_1}}{\mathrm{r} - \mu_1} - \frac{\mathrm{QP_2}}{\mathrm{r} - \mu_2} - \frac{\mathrm{C_{o\&m}}}{\mathrm{r}} \end{aligned} \tag{8}$$

Investment option value

When investment has not been made, as [I] = 0, and there is no revenue flow, we can consider the investment option whose value can be defined as $F(P_1,\ P_2)$. According to the Bellman equation, the rate of return on the option should equal the expected rate of capital gain from it, which is shown in equation (9) (Dixit and Pindyck[24]) This equation is necessary to find the trigger prices of $P_1,\ P_2$ from the free boundary condition above which the investment becomes optimal.

$$rF(P_1,\ P_2) = \frac{1}{d_t}E_0[dF(P_1, P_2)] \tag{9}$$

By Ito's lemma, $dF(P_1,\ P_2)$ can be acquired as equation (10), where the subscript denotes partial differentiation, $F_i = \partial F(P_1, P_2)/\partial P_i$, $F_{ij} = \partial F(P_1, P_2)/\partial P_i \partial P_j$ for {i, j = 1', 2}.

$$dF(P_1,\ P_2) = F_1 dP_1 + \frac{1}{2}F_{11}dP_1^2 + F_2 dP_2 + \frac{1}{2}F_{22}dP_2^2 + F_{12}dP_1P_2 \tag{10}$$

By taking expectation to equation (10), we can get equation (11) based on properties of

$$\begin{aligned} &E_0[dt]^2 = 0,\ \ E_0[dtdw] = 0,\ \ E_0[dw]^2 = dt \text{ and } Cov[dP_1, dP_2] = \rho_{12}\sigma_1\sigma_2 dt;\\ &\ \rho_{12} \in [-1,1]\frac{1}{dt}E_0[dF(P_1, P_2)]\\ &= \left[F_1\mu_1P_1 + \frac{1}{2}F_{11}\sigma_1^2P_1^2 + F_2\mu_2P_2 + \frac{1}{2}F_{22}\sigma_2^2P_2^2 + \rho_{12}\sigma_1\sigma_2P_1P_2F_{12}\right]dt \end{aligned} \tag{11}$$

By plugging equation (11) into equation (9), we can further derive the Bellman equation as (12)

$$rF(P_1,\ P_2) = F_1\mu_1P_1 + \frac{1}{2}F_{11}\sigma_1^2P_1^2 + F_2\mu_2P_2 + \frac{1}{2}F_{22}\sigma_2^2P_2^2 + \rho_{12}\sigma_1\sigma_2P_1P_2F_{12} \tag{12}$$

The general solution to the partial differential equation has the power form shown in equation (13). (Adkins and Paxson, (2008)). A, $\beta,\ \eta$ are coefficients that need to be determined with an optimal investment price of P_1^* depending on P_2.

$$F(P_1,\ P_2) = AP_1^{\beta}P_2^{\eta}; 0 < P_2 < \infty,\ \ 0 < P_1 < P_1^*(P_2) \tag{13}$$

Equation (13) can be further developed to equation (14) by different signs that β and η can have as equation (15)

$$F(P_1,\ P_2) = A_1P_1^{\beta_1}P_2^{\eta_1} + A_2P_1^{\beta_2}P_2^{\eta_2} + A_3P_1^{\beta_3}P_2^{\eta_3} + A_4P_1^{\beta_4}P_2^{\eta_4} \tag{14}$$

where

$$\beta_1 > 0,\ \ \eta_1 < 0$$

$$\beta_2 < 0,\ \ \eta_2 > 0$$

$$\beta_3 > 0,\ \ \eta_3 > 0$$

$$\beta_4 < 0,\ \ \eta_4 < 0 \tag{15}$$

Equation (14) can be simplified by a limiting boundary condition on P_1 and P_2. When P_2 converges to infinity, the option value converges to zero, because the Gap price generates cost flows in this investment; so A_2 and A_3 should be zero to get rid of this possibility. In contrast, the REC price generates important revenue flows in this investment; so if it converges to zero, investment in storage projects cannot be justified, and the option value becomes zero. Hence, A_4 should be also zero. Then, A_1 with $\beta_1 > 0,\ \eta_1 < 0$ is the only feasible part of the equation where investment is justified; so we can now simplify equations (14) to (16)

$$F(P_1, P_2) = AP_1^{\beta}P_2^{\eta},\ \ 0 < P_2 < \infty,\ \ 0 < P_1 < P_1^*(P_2), \beta\ >\ 0,\ \eta\ <\ 0 \tag{16}$$

Calculating optimal investment prices

Now we can use two boundary conditions, which are the value-matching condition and the smooth-pasting condition, shown in equations (17) and (18), to find the unknown variables A, β, and η in equation (16). The optimal investment price of P_1 and P_2 that makes an investment valid will be determined once those unknown variables are estimated (Dixit and Pindyck[24])

Value-matching condition:

$$V(P_1,\ P_2) = F(P_1,\ P_2) \quad \text{at } P_1 = P_1^*(P_2) \tag{17}$$

Smooth-pasting condition:

$$\frac{\partial V(P_1^*,\ P_2^*)}{\partial P_i^*} = \frac{\partial F(P_1^*,\ P_2^*)}{\partial P_i^*}, \quad \text{at } P_1 = P_1^*(P_2), \tag{18}$$

where I = 1, 2

Equation (19) shows the value-matching condition that illustrates the expected project value, and the investment option value should be the same as the optimal investment price.

$$\frac{\mathrm{nEQP_1}}{\mathrm{r}-\mu_1}-\frac{\mathrm{QP_2}}{\mathrm{r}-\mu_2}-\frac{\mathrm{C_{o\&m}}}{\mathrm{r}}=\mathrm{AP_1^{\beta}P_2^{\eta}},\quad \text{at } P_1=P_1^*(P_2) \tag{19}$$

Equations (20) and (21) show the smooth-pasting condition, which means that the project value and investment option should meet continuously and tangentially at optimal investment prices. We have two prices that the optimal investment region needs in order to be determined; so the smooth-pasting conditions are presented in two equations based on the tangential condition for REC price and GAP price, respectively.

$$\frac{\mathrm{nEQ}}{\mathrm{r}-\mu_1}=\beta\mathrm{AP_1^{\beta-1}P_2^{\eta}},\quad \text{at } P_1=P_1^*(P_2) \tag{20}$$

$$-\frac{\mathrm{Q}}{\mathrm{r}-\mu_2}=\eta\mathrm{AP_1^{\beta}P_2^{\eta-1}},\quad \text{at } P_1=P_1^*(P_2) \tag{21}$$

Rearranging equation (20), we can obtain the endogenous coefficient, A, as follows in equation (22)

$$\mathrm{A}=\frac{\mathrm{nEQ}}{\beta(\mathrm{r}-\mu_1)}\mathrm{P_1^{*1-\beta}P_2^{-\eta}} \tag{22}$$

Plugging equation (22) into equation (21), the optimal stopping boundary, $\mathrm{P_1^*(P_2)}$, can be organized as shown in equation (23)

$$\mathrm{P_1^*(P_2)}=\frac{\beta(\mathrm{r}-\mu_1)\mathrm{P_2}}{\eta\mathrm{nE}(\mu_2-\mathrm{r})} \tag{23}$$

Combining equation (23) and equation (19) gives us the endogenous coefficient, η, as follows in equation (24)

$$\eta=\frac{\mathrm{rQP_2}(1-\beta)}{\mathrm{rQP_2}+\mathrm{C_{o\&m}}(\mathrm{r}-\mu_2)+\mathrm{rI}(\mathrm{r}-\mu_2)} \tag{24}$$

By combining equations (12), (16), and (24), we can get rid of A and η, and finally obtain the quadratic polynomial equation with β only as shown in equation (25)

$$\mathrm{a}\beta^2+\mathrm{b}\beta+\mathrm{c}=0, \tag{25}$$

where

$$\mathrm{a}=\frac{1}{2}\sigma_1^2+\frac{1}{2}\sigma_2^2\mathrm{H}^2-\rho\sigma_1\sigma_2\mathrm{H} \tag{26}$$

$$b = \frac{1}{2}\sigma_2^2 H^2 - \sigma_2^2 H - \frac{1}{2}\sigma_1^2 + \mu_1 - \mu_2 H + \rho\sigma_1\sigma_2 H \tag{27}$$

$$c = \frac{1}{2}\sigma_{GAP}^2 H^2 - \frac{1}{2}\sigma_{GAP}^2 H + \mu_{GAP} H - r \tag{28}$$

where

$$H = \frac{rQP_2}{rQP_2 + C_{o\&m}(r - \mu_2) + rI(r - \mu_2)} \tag{29}$$

The discriminant is $\Delta = -b \pm \sqrt{b^2 - 4ac}$; therefore, we can ensure the existence of two real and distinct roots. We need to take a positive value for β in equation (30) to ensure that the REC price will not diverge, as we discussed to determine the condition in equation (16)

$$\beta_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} > 1 \tag{30}$$

$$\beta_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} < 0 \tag{31}$$

Finally, we can simplify the option value function, equation (16), by replacing A from equation (22) and η from equation (24), so that it becomes equation (32)

$$F^*(P_1, P_2) = \frac{nEQ}{\beta(r - \mu_1)} P_1^{*1-\beta} P_2^{\eta}, \ 0 < P_2 < \infty, \ 0 < P_1 < P_1^*(P_2) \tag{32}$$

The free boundary for REC price in equation (33) can be found by combining equation (23), β from equation (30), and η from equation (24). This equation for the optimal investment price for P_1, given P_2, is the main result of this study.

$$P_1^*(P_2) = \frac{\beta(r - \mu_1)[rQP_2 + C_{o\&m}(r - \mu_2) + rI(r - \mu_2)]}{rnEQ(r - \mu_2)(\beta - 1)} \tag{33}$$

$P_1^*(P_2)$ in equation (33) means that, given P_2, it is optimal to invest if $P_1 > P_1^*$, but not if $P_1 < P_1^*$. If we apply the same derivation process for P_2 given P_1, we can arrive at equation (34).

$$P_2^*(P_1) = \frac{\eta(r - \mu_1)[rnEQP_2 - (r - \mu_2) - rI(r - \mu_2)]}{rQ(\eta - 1)(r - \mu_2)} \tag{34}$$

Both the REC price and the GAP price are volatile and affect the optimal investment decision. However, it turns out that the REC price has a more significant effect on NPV and option value, as shown in the numerical example provided in the next section. Hence, we focus on analyzing the optimal investment price of REC based on equation (33).[25]

Numerical example

Data

Table 1 summarizes the market and technical data for the numerical example of this study. It is based on the actual project implemented by LG, and this benchmark project consists of an energy-storage unit whose power capacity is 1 MW and energy capacity is 2.88 MWh. The market data applied in this study is as of June 2019.

Q is the annual quantity of electricity purchased using the storage by the benchmark project, which is computed as 2.88 MWh (energy capacity) times 365 (days). $P_{REC,\ 0}$ and $P_{GAP,\ 0}$ show current REC price and Gap price level, which are 59.79 USD/MWh and 7.93 USD/MWh, respectively. 90% of storage efficiency is applied, and project overnight investment cost and operation and management cost are from the benchmark project of LG, which is one of the major developer with many project records in Korea. To see more clear

Table 1. Market and technical parameter values.

Model symbol	Description	Value	unit
Q	Annual quantity of electricity purchased from the market[a]	1022	MWh/year
$P_{REC,\ 0}$	Annual average Renewable Energy Certificate price[b]	59.79	USD/MWh
$P_{buy,\ 0}$	Annual average buying price from market at 10 am to 4 pm[b]	81.43	USD/MWh
$P_{sell,\ 0}$	Annual average selling price from market at 5 pm to 24 pm[b]	81.86	USD/MWh
$P_{GAP,\ 0} = P_{buy} - E \cdot P_{sell}$	Price gap between charging price and discharging price[b]	7.93	USD/MWh
E	Energy storage conversion efficiency[c]	90	%
$C_{o\&m}$	Operating and maintaining cost[c]	221.74	thousand USD
I	Investment cost[c]	1,389	thousand USD
n	Number of RECs per MWh in 2020[b]	4	
r	Discount rate	0.045	
μ_{REC}	Drift rate of REC[b]	–0.0567	
σ_{REC}	Volatility rate of REC[b]	0.1175	
μ_{GAP}	Drift rate of Gap price[b]	–0.0335	
σ_{GAP}	Volatility rate of Gap price[b]	0.3212	
ρ	Correlation coefficient[b]	–0.309	

[a]Based on storage power capacity of 1 MW and storage energy capacity of 2.88 MWh from the actual project data (LG).
[b]Korea Power Exchange (KPX) database, epsis.kpx.or.kr.
[c]Technical specification of storage from project data (LG).

effects in 2020, we used an REC multiplier of four, which is the number that the RPS scheme in Korea applies for energy- storage in 2020. Drift rate and volatility of the REC price and the GAP price are specified, and the correlation coefficient between the two prices is also presented in Table 1.

Optimal boundary condition

Figure 2 shows the optimal boundary condition of the option value and project value based on REC price. This figure shows how the value-matching condition and smooth-pasting condition are satisfied between the expected net present value (NPV) of the project and the option value at the optimal trigger price of the REP price. The optimal boundary condition of the REC price was presented as a function of the GAP price in this figure, as noted in the model description, so this figure displays how the boundary condition of the REC price is determined at the current GAP price. Under NPV, the optimal trigger price of investment is 50.34 USD/MWh, where the NPV line crosses the zero line in the project value, and under option value, the optimal trigger price is estimated as 56.27 USD/MWh, where option value and NPV meet, which is approximately 11.8% higher. This means that the optimal trigger price based on option value is determined when the waiting value in the option value is fully compensated for by the NPV, and no more waiting value is left. In addition, the option value cannot go negative and is always above the project NPV, because the value of the option to choose whether to invest or not cannot be negative, since there is always an option to give up the project when it is not profitable. The option value can be considered as the sum of fundamental project values and the option to choose in or out, so it should be always higher than or equal to the project NPV.

Table 2 summarizes the optimal investment price of the REC price under NPV and real option, and shows the option value and project value under the expected NPV at the current prices of REC and GAP. At the current REC price, the trigger price under NPV is satisfied but that under option value is not fulfilled, which indicates that the current REC price level is not high enough to compensate for the project risk caused by price volatilities. The project NPV at current prices is estimated as 3,79,919 USD, and the option value is estimated as 3,40,904 USD.

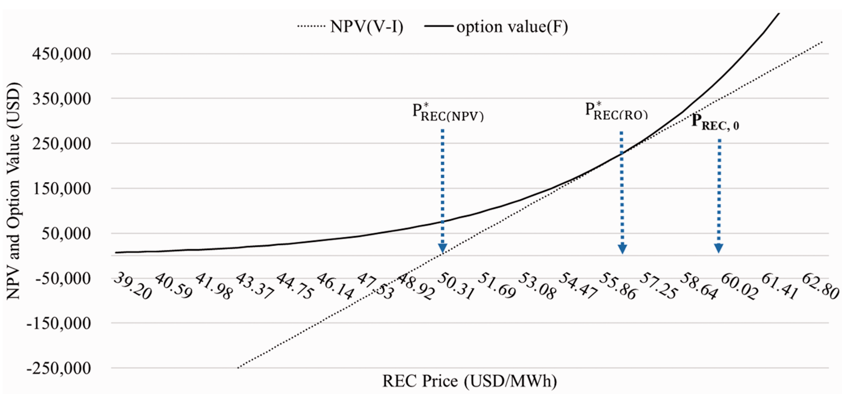

Figure 2. Optimal investment price of investment under NPV and option value.

Comparing with results from previous literatures that applied real option methodology, Heydari et al.[4] shows that option value of partial carbon capture and storage(PCCS) under volatilities of carbon price and electricity price is approximately 47.7% higher compared to net present value(NPV), and that of full carbon capture and storage(FCCS) is approximately 404.5% higher compared to NPV. Zhang et al.[26] uses American option model to analyze the investment decision of solar pv project by considering uncertainties of carbon price and non-renewable energy(NRE) cost and showed that the difference between real option value and NPV is negligible due to small volatility in carbon price and NRE cost.

Figure 3 shows the optimal stopping boundary condition for $P_{REC}*(P_{GAP})$. The line for the optimal stopping boundary can be interpreted as meaning that, if the REC price is located above the line for a given GAP price, the investment is justified right away, but if it is below the line, the investment is not valid, and we say it is in the waiting area. As expected, when the GAP price increases, we want a higher REC price to make the investment more profitable, since the GAP price works as a cost component in this investment.

The optimal stopping boundary condition for the REC price has the form shown in equation (35) as a function of the GAP price. This equation shows that even though the GAP price is zero, the project still requires an REC price of 53.083 USD/MWh to make the project feasible under price volatilities. Also, this equation shows that when the GAP price increases by 1 USD, the REC price should be increased by 0.4022 USD to make the investment feasible. This means that a 1 USD change in the GAP price is equivalent to a 0.4022

Table 2. Parameter Simulation Value.

P^{*}_{REC}	$P^{*}_{REC(NPV)}$	P^{0}_{REC}	F_0	V_0-I
56.27 USD/MWh	50.34 USD/MWh	59.79 USD/MWh	379,919 USD	340,904 USD

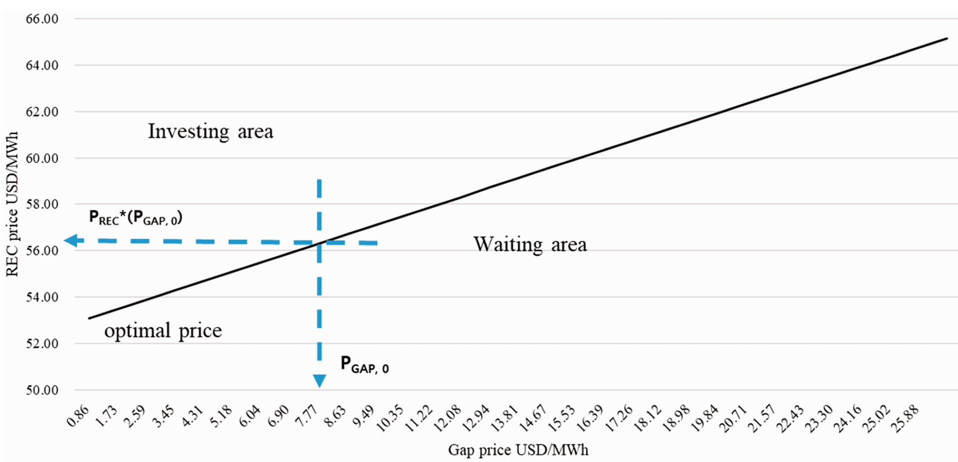

Figure 3. Optimal stopping boundary condition for $P_{REC}*(P_{GAP})$.
(a) P_I^* under option value and NPV (b) F_0 and V_0-I

USD change in the REC price, which shows that the REC price is the more dominant factor determining the optimal investment timing.

$$P^*_{REC}(P_{GAP}) = 0.4022 \cdot P_{GAP} + 53.083 \tag{35}$$

Impact of varying REC multipliers

Figure 4 shows how the different REC multipliers for energy storage affect the optimal investment price, project value, and option value. Under the current RPS scheme in Korea, the number of RECs offered to energy storage per MWh of energy delivered decreased from 5 in 2019 to 4 in 2020, as the cost of a lithium-ion battery(LIB) is projected to go down and to go further down to 3 in the near future. Knowing that the REC is an important revenue source for storage investment, it is important to verify the effect of decreasing the REC multiplier on the optimal investment price and project value. We found that when the REC multipliers decrease from 5 to 3, the optimal investment price of the REC under option value decreases from 75.03 to 45.02 USD/MWh and the optimal investment price under NPV decreases from 67.12 USD/MWh to 40.27 USD/MWh. Compared to the optimal investment price of 56.27 USD/MWh under a 4 REC multiplier, the 3 REC multiplier and 5 REC multiplier require a 33.3% higher optimal price and a 20.0% lower optimal price to make the investment feasible.

For values, it is shown that the REC multiplier has significant effects on both option value and project NPV. When the REC multiplier decreases from 5 to 3, the option value decreases from 3,155 thousand USD to 25 thousand USD, and NPV decreases from 881 thousand USD to -200 thousand USD. The option value increases exponentially as the REC multiplier increases and NPV increases linearly, which shows that the investment is not feasible under a 3 REC multiplier at the current price levels and cost structure. As seen in Figure 4, the REC multiplier has a significant effect on the optimal investment price and project value; knowing this, the current RPS scheme should be used carefully when further reducing the REC multiplier for energy storage.

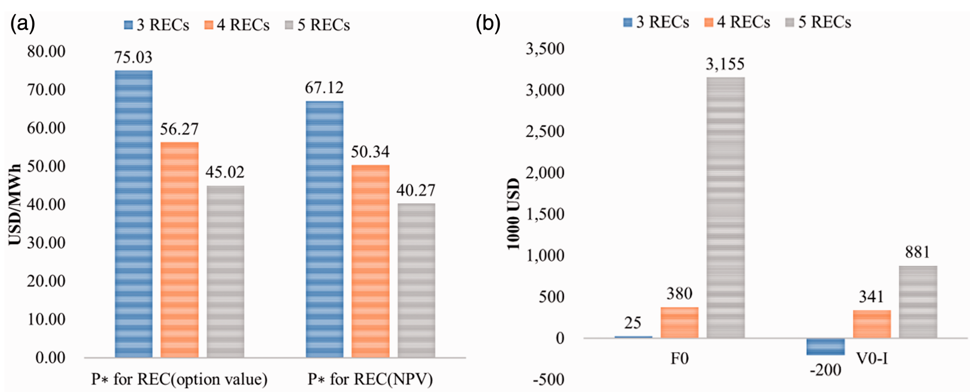

Figure 4. Optimal investment prices and project values under various REC multipliers.

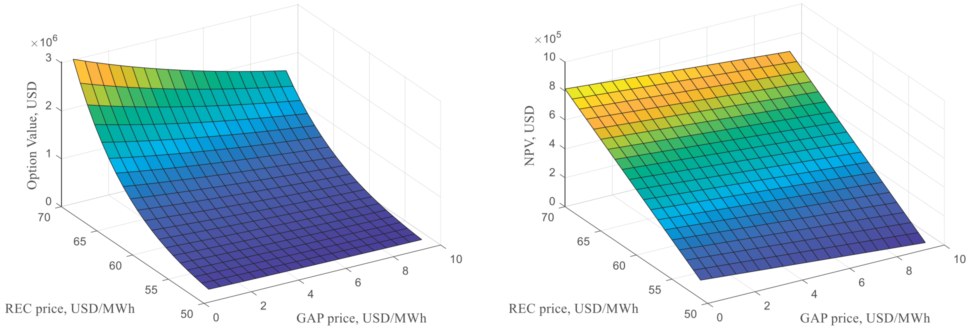

Figure 5. Sensitivity of option value and NPV to REC price and GAP price.

Sensitivity analysis

Figure 5 shows how the option value and NPV are affected by two price levels. When the REC price changes from 50 USD/MWh to 70 USD/MWh with a GAP price fixed at zero, the option value changes from 132 thousand USD to 3081 thousand USD. When the GAP price changes from 0 USD/MWh to 10 USD/MWh with the REC price fixed at 50 USD/MWh, the option value change from 74 thousand USD to 132 thousand USD. This shows that option value is much more sensitive to REC price change. The sensitivity of NPV to REC price and GAP price shows results similar to the those of the option value case. In addition, the increase of option value because of REC price increase is non-linear; the reason for this can be confirmed by Figure 2, which shows that the option value function has a convex form.

Figure 6 shows how the optimal investment price of REC under the option value changes for various drift rates and volatilities when the drift rate and volatility of the GAP price is fixed at the current level. The figure shows that when the volatility of the REC price is zero, the current level is (0.1175), and the doubled level is (0.235), the optimal investment prices are estimated as 51.35 USD/MWh, 56.27 USD/MWh, and 71.23 USD/MWh, respectively, at the current drift rate. This shows that the optimal investment price increases by 26.6% when volatility of the REC price gets doubled from the current level and decreases by 8.7% when the volatility of the REC price is completely eliminated. This result emphasizes that price volatility has a significant effect on investment decisions on energy storage, and indicates that the financial burden for the government to support the RPS can be greatly reduced if the volatility of the REC price is effectively reduced.

The number of RECs issued in 2018 under the RPS scheme in Korea was 22,885,601. This means that the cost the government has to bear to support the RPS is approximately 1368 million USD; if the REC price volatility is eliminated, 119.6 million USD can be saved and used to support additional deployment of storage and renewable sources.

In order to promote energy storage to support the power grid with high penetration of various renewable sources, a strong policy that can reduce the volatility of the REC price is necessary to reduce the burden for both investors and government in order to achieve the renewable capacity target more effectively.

Figure 7 shows the sensitivity of the optimal trigger price of the REC to LIB capital cost under RPS and FIT. The dash line shows the level of current LIB capital. This result shows that when LIB capital cost reaches 10,00,000 USD, the optimal REC price goes down to

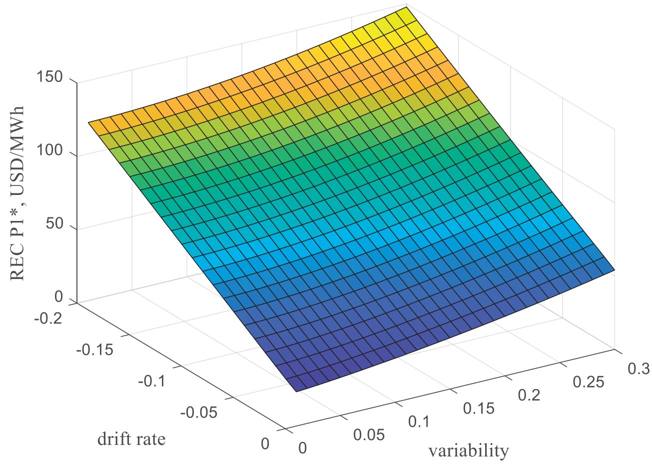

Figure 6. Sensitivity of the optimal trigger price of the REC to drift rate and volatility.

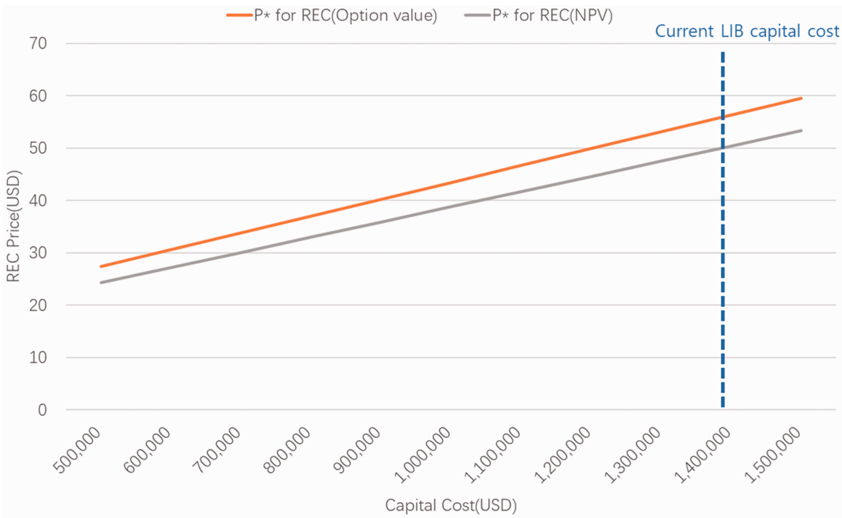

Figure 7. Sensitivity of the optimal trigger price of the REC to LIB capital cost.

43.4 USD/MWh and 38.8 USD/MWh for option value and NPV, respectively. When LIB capital cost becomes half of the current cost which is approximately 7,00,000 USD, the optimal price becomes 33.8 USD and 30.1 USD, respectively.

Conclusion and policy implication

This study analyzed how the multiple price volatilities under the RPS scheme affect the investment decisions on energy storage that is essential to reliably operate the power system with a high penetration of solar and wind generation. We applied the real-option methodology and considered the stochastic processes of the REC price and GAP price when deriving optimal investment prices and project values under option value and NPV.

We found that the REC price has a more dominant effect on investment decisions and project value than does the GAP price because of a high REC multiplier. The reduction of the REC multiplier to 4 in 2020 significantly increased the optimal investment price of the REC by 20.0%. With the current volatility level of the REC price and GAP price, the optimal REC price for investment under option value and NPV was 56.27 USD/MWh and 50.34 USD/MWh, respectively, which shows that the current multiple price volatilities increase the optimal investment price by 10.5%.

This result gives us following policy implications. First, the government needs to focus on reducing the volatility of the REC price in order to provide better incentives for project investment. The high REC multiplier and high price level of the REC make it a more important factor in determining the optimal investment price of an energy-storage project.

Second, a more stable subsidy scheme, such as FIT, needs to be reconsidered in order to attract more projects of energy storage at relatively lower REC prices. We have shown that the optimal trigger price of REC under no volatility is 51.35 USD/MWh, that with the current volatility level of 0.1175 is 56.27 USD/MWh, and that with a doubled volatility level of 0.235 is 71.23 USD/MWh. The high volatility of the REC price is a crucial factor affecting the optimal trigger price and investment decisions. Under the RPS scheme, financial expense caused by the REC payment is a burden for the government and eventually for customers, because this subsidy payment must be reflected in the electricity price. Hence the FIT scheme, which has no volatility in the REC price, has a much lower financial burden than the RPS scheme has in order to achieve the same level of energy-storage deployment.

Third, a new subsidy policy, such as an auction, needs to be considered for more efficient deployment of not only energy-storage but also other renewable sources. The advantage of RPS is that it brings the market competition mechanism among the investments of renewable energy, so it helps reduce the project cost, but the disadvantage is, as shown in this study, that the volatility of the REC price increases the optimal trigger price and discourages investment, whereas the advantage of FIT is that it eliminate the volatility of the REC price and provides a stable subsidy, but since it does not use the market mechanism, finding the proper level of subsidy payment is difficult, and when it is over-estimated, the financial burden for the government increases and the motivation to reduce the project cost decreases.

The solution that can accommodate the advantages of the two programs is an auction, which provides a stable payment to energy storage projects or other renewable energy projects for contracted periods, and the payment is determined by the bidding process. Hence, in this auction scheme, searching for the correct subsidy level is done by the competitive market mechanism, which is the advantage of the RPS. Once the subsidy level is determined, this payment is stably made to investors, which is the advantage of FIT. In addition, in the bidding process, a surplus of project providers is transferred to electricity customers, since the auction contract is settled at the minimum willingness-to-accept price for investors. Hence, the auction scheme will be able to reduce the financial burden of the subsidy policy significantly by eliminating subsidy volatility and finding the minimum willingness-to-accept from investors. In addition, investors can lower the financing cost under auction scheme as the uncertainties in SMP and REC price are eliminated.

As global consensus has made about transition to a low-carbon economy, various subsidy programs have been adopted in order to rapidly deploy various renewable sources and the energy-storage capacities that help improve the efficiency of those sources. The main goal here is to achieve the deployment target within the limited financial burden that will be eventually transferred to customers. In order to achieve this goal, we will have to come up with a sophisticated policy that can attract more investments with a lower subsidy.

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